

Trackformer 1.2

Coupled pressure-field forecasting

Architecture, forecast dynamics and development evaluation

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[Code, data and model release](#)

Abstract

Trackformer 1.2 forecasts tropical-cyclone motion and central pressure by jointly advancing a basin-scale atmospheric state and a translating pressure core. Recurrent convolutional networks evolve the fields, while multiscale attention supplies environmental context. A pressure-associated centre and central-pressure sample are read from the evolving core at each six-hour step. Twenty autoregressive transitions produce a five-day forecast. This report specifies the released architecture, state updates, learning objectives and 50-member input-perturbation policy. Across 1,473 daily forecasts from 270 storms, equal-storm mean track error is 471.2 km for the 1.2 ensemble and 798.4 km for the causal 1.1 route pipeline. The pipelines differ; this is not an architecture ablation or an untouched-test result. A selected Fung-wong case illustrates actual geographic alignment, intensity evolution and model-generated pressure contours, including their remaining errors and coverage limits.

Forecast horizon	Representation	Evaluation policy
+6 to +120 h	Basin + moving pressure core	50 input realizations
20 six-hour transitions	21.45 million parameters	One shared set of weights

1 Overview and forecast state

The defining constraint is a coupled readout: the route and central pressure must follow the pressure core generated by the model. A learned, steering-conditioned proposal moves the core’s reference frame; a local pressure association then determines the reported centre. The method is therefore neither a pressure-only tracker nor a complete numerical atmospheric solver. Its displayed pressure is a forecast field, not an independently drawn vortex placed along a predicted route.

At step k , the recurrent state is

$$s_k = (g_k, h_k^g, c_k, h_k^c, a_k, o_k, z_k, m_k, v_k; G, R, I, M).$$

State	Meaning
g, h^g	Eight normalized basin fields and basin memory.
c, h^c, a	Core pressure, core memory and pressure anomaly relative to the basin.
o, z, m, v	Core-grid origin, associated centre, six-hour motion and auxiliary wind.
G, R, I, M	Fixed geography, regional coordinates, issue intensity and validity masks.

Scope. A research model for the Western Pacific, not an operational warning system. Selected examples do not establish generalization or safety.

2 Architecture and input contract

The basin and moving core share a recurrent forecast cycle, but retain separate spatial representations and memories. Figure 1 shows the dependency structure; the operator definitions follow in Section 3.

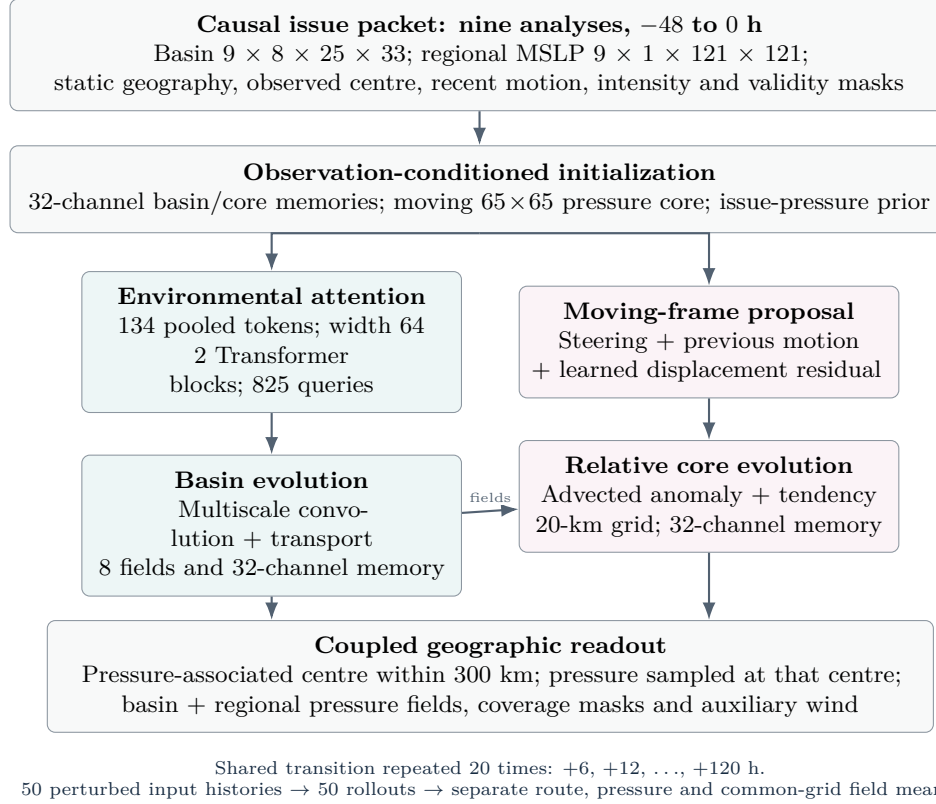


Figure 1. Coupled forecast architecture. Basin evolution supplies environmental fields to the moving core; the pressure-associated readout follows the motion proposal. All leads and ensemble members use the same trained network.

2.1 Causal inputs and spatial representations

Input / state	Shape / sampling	Role and boundary
Basin history	$B \times 9 \times 8 \times 25 \times 33$	Nine analyses at $-48, -42, \dots, 0$ h; 0 – 60°N , 100 – 180°E , 2.5° spacing.
Regional history	$B \times 9 \times 1 \times 121 \times 121$	Issue-anchored MSLP patch, $\pm 15^\circ$, sampled every 0.25° ; availability mask selects native or coarse initialization.
Moving core	65×65 ; 20-km spacing	Learned pressure reconstruction over ± 640 km; grid spacing does not establish effective resolution.
Issue state	Centre, motion, intensity, masks	Latitude/longitude, six-hour east/north displacement, wind in knots, pressure in hPa.
Static channels	Four channels per grid	Latitude/90, longitude/180–1, land fraction and elevation/8000 m.

Basin channels are $[\text{MSLP}, Z_{500}, u_{850}, v_{850}, u_{500}, v_{500}, u_{200}, v_{200}]$ in hPa, metres and metres per second. Normalization uses fixed training-only statistics, $\tilde{x}_j = (x_j - \mu_j)/\sigma_j$. Regional coordinates remain available when native history is missing. Only information available at or before issue time may enter inference; retrospective archive coverage does not establish operational availability. Outputs include basin and regional pressure, core coordinates, centre, central pressure, masks and auxiliary wind; there are no quadrant wind radii or resolved surface-wind maps.

3 Neural operators

The released model has 21,452,595 trainable parameters.

3.1 Multiresolution evolution networks

The basin network receives 44 channels: eight fields, 32 memory features and four static channels. Encoder widths are 72, 144, 288, 432; residual-block depths are 2, 2, 4, 4. A 3×3 stem and stride-two convolutions produce four scales. The decoder uses 1×1 projections, bilinear upsampling, additive skip connections and one residual block at each reconstructed scale. Output heads emit ten transition channels and 32 recurrent features. Eight transition channels are field tendencies and two are flow corrections.

Each residual block comprises a depthwise 5×5 convolution, eight-group normalization, 1×1 expansion by four, GELU and 1×1 contraction. A learned residual multiplier starts at 0.01. Transition heads start at zero. The core network uses widths 64, 128, 256, 384 and the same depths. Its 45 inputs are core pressure, eight environmental fields, 32 core-memory channels and four geographic channels. Three outputs represent one anomaly tendency and two local-flow corrections.

3.2 Convolutional recurrent memory

Basin and core memories each have 32 channels. History encoders receive 72 flattened basin channels or nine pressure-history channels, respectively. Given advected memory h and new evolution features f ,

$$\begin{aligned}(r, u) &= \text{sigmoid}(W_g * [h, f]), \\ q &= \tanh(W_q * [r \odot h, f]), \\ h' &= (1 - u) \odot h + u \odot q.\end{aligned}$$

The gate convolution emits 64 channels split into reset and update gates; the candidate convolution emits 32. Forecast leads share these recurrent modules rather than using twenty independent predictors.

3.3 Environmental attention

A 1×1 convolution projects $[g, h^g, G]$ from 44 to 64 channels. Average pooling at sizes/strides three and five with ceiling mode yields $9 \times 11 + 5 \times 7 = 134$ environmental tokens. Two pre-normalization Transformer layers use width 64, four heads, feed-forward width 128, ReLU activation and zero dropout. Each of the 825 basin cells supplies a query projected from its memory to width 64. Four-head cross-attention retrieves context; a zero-initialized $64 \rightarrow 32$ projection adds it to basin memory before the transition. No separate positional embedding is added: geographic coordinates are already among the token inputs.

3.4 Observation-conditioned initialization

Write $\mathcal{S}(F, \xi)$ for bilinear field sampling. Sampling uses FP32 geometry, aligned corners and border padding for the basin; transported core anomalies use zero padding. All nine pressure histories are sampled onto the core grid centred at the issue observation. A detail mask chooses native pressure or coarse pressure. The latest map initializes c_0 , while all nine maps initialize the core memory.

If issue central pressure is valid, the initial core is corrected by

$$c_0(x, y) \leftarrow c_0(x, y) + (\tilde{p}_{\text{issue}} - c_0(0, 0))e^{-(x^2+y^2)/(2 \cdot 100^2)}T(x, y),$$

where x, y are kilometres and $T = \text{clip}((640 - \sqrt{x^2 + y^2})/160, 0, 1)$. The anomaly is $a_0 = c_0 - \mathcal{S}(g_{0,\text{MSLP}}, \xi_0)$. This compact prior is applied only at initialization. Future scalar pressure is not repeatedly inserted into the field.

4 One six-hour forecast transition

Each step advances the basin first, proposes core motion, transports the core relative to that motion, and reads out position and pressure. All twenty leads share the same operators. Sampling $\mathcal{S}(F, \xi)$ is bilinear; coordinates and units are explicit below.

4.1 Environmental transport

Learned softmax weights blend 850-, 500- and 200-hPa winds; their initial values are (0.4, 0.4, 0.2). For basin-network output d , the corrected flow and next normalized basin field are

$$U_k = \sum_{\ell} \alpha_{\ell} (u_{\ell}, v_{\ell})_k + 15 \tanh(d_{8:10}), \quad (1)$$

$$g_{k+1} = \mathcal{S}(g_k, D(U_k)) + 0.2 \tanh(d_{0:8}). \quad (2)$$

U_k is in ms^{-1} and D is the backtraced departure grid. Six hours convert 1 ms^{-1} to 21.6 km. Coordinate conversion uses 111.2 km per latitude degree and $111.2 \max(\cos \phi, 0.2)$ km per longitude degree. The attention-conditioned memory follows the same advection and then a gated update. This transport is not an exactly conservative atmospheric solver.

4.2 Moving-frame proposal

A $44 \rightarrow 64 \rightarrow 3$ GELU MLP reads 32 sampled memory channels, eight basin fields, previous motion divided by 100, and two issue-intensity masks. For output b ,

$$m_{k+1} = 0.7(21.6)U_k(z_k) + 0.3m_k + 100 \tanh(b_{0:2}). \quad (3)$$

This km-per-six-hour displacement defines the proposed origin relative to z_k , not the final reported centre. Auxiliary wind becomes $v_{k+1} = \max(0, v_k + 5b_2)$.

4.3 Core transport in the moving frame

The core network reads previous core pressure, the old sampled environment, memory and geography at the proposed origin. Its local velocity is $U^{\text{local}} = \mathcal{S}(U_k) + 5 \tanh(d_{1:3}^c)$. Subtracting $m_{k+1}/21.6$ removes bulk frame motion. The relative departure grid D_c also accounts for the previous centre–origin offset:

$$a_{k+1} = \mathcal{S}(a_k, D_c) + 0.1d_0^c, \quad (4)$$

$$c_{k+1} = \mathcal{S}(g_{k+1, \text{MSLP}}, \xi_{k+1}) + Ta_{k+1}. \quad (5)$$

The anomaly tendency d_0^c is unbounded by $\tanh$. Core memory uses the same relative sampling followed by its gated update.

4.4 Pressure-associated readout

Convert core pressure to hPa, $P = \sigma_p c_{k+1} + \mu_p$. For cells no more than 300 km from the proposed origin, define

$$w_i = \text{softmax}_i(-P_i/2 - r_i^2/(2 \cdot 180^2)). \quad (6)$$

The weighted mean of cell latitude/longitude gives z_{k+1} ; central pressure is $\mathcal{S}(P, z_{k+1})$, not necessarily the grid minimum. Cells outside the radius are excluded, reducing association with unrelated depressions.

4.5 Geographic pressure reconstruction

The issue-anchored regional field is

$$\hat{p}_R = \mathcal{S}(g_{k+1, \text{MSLP}}, R) + \mathcal{S}(Ta_{k+1}, R; \text{zero padding}), \quad (7)$$

then de-normalized to hPa. This is the model’s basin-plus-core field; future observations or display-only vortices are never substituted. Coverage masks must remain visible. Resampling and local centre association mean the route, central pressure and displayed field minimum need not coincide exactly.

5 Training, ensembles and reproducibility

5.1 Autoregression and learning objectives

Inference initializes once and repeats the transition twenty times, using only fixed issue context and its own predicted future state. The implemented single-step loss is

$$L = L_{\text{env}} + L_{\text{regional}} + 0.2L_{\text{route}} + 0.5L_{\text{pressure}} + 0.05L_{\text{wind}} \\ + 0.25L_{\Delta p} + 0.1L_{\Delta z}.$$

The final terms require valid preceding-target labels. Environment loss is cosine-latitude-weighted Smooth L1 in normalized units. Regional loss includes Smooth L1 and 0.2 times adjacent-grid gradient errors on each axis. Absolute regional-field supervision excludes the inner 150 km around the target centre, because reanalysis can underresolve observed central pressure. This is a supervision choice, not proof of correct inner-core structure.

Route residuals use a 100-km scale; pressure and wind use 20 hPa and 30 kt. Pressure-change error uses 10 hPa and displacement-change error uses 100 km. All objectives use explicit masks. They train field evolution and readout together, without adding an independent central-pressure output head. Training-only normalization, storm-level split boundaries and causal input selection remain essential implementation constraints.

5.2 Single forecasts and 50-member means

Evaluation-mode inference is deterministic for fixed inputs and weights. The documented ensemble perturbs only normalized historical weather: independent Gaussian fields are spatially smoothed with kernels five (basin) and nine (regional), centred, unit-scaled per map and multiplied by amplitudes 0.025 and 0.015. Issue centre, motion, intensity, masks and geography remain fixed. Each member performs a complete recurrent rollout.

With $M = 50$, routes and central pressures are averaged separately: $\bar{z} = M^{-1} \sum_m z^{(m)}$ and $\bar{p} = M^{-1} \sum_m p^{(m)}$. Physical pressure fields are averaged on common geographic grids, not by averaging unaligned moving-core array indices. In general,

$$\bar{z} \neq \arg \min_x \bar{P}(x), \quad \bar{p} \neq \min_x \bar{P}(x).$$

These are perturbed-input realizations of one model, not 50 trained networks or learned latent samples. Spread is not automatically calibrated uncertainty. Member counts, perturbation settings and seed policy must accompany the outputs.

5.3 Release identity and reproducibility

The public model is **Trackformer 1.2**; the manifest retains internal checkpoint identity 1.2.73 epoch 4 and exact source/weight hashes. The following artifacts are distributed in the [release repository](#):

Artifact	Purpose
<code>model.py</code>	Environmental attention and additional temporal losses.
<code>baseline_model.py</code>	Moving-core evolution and pressure-associated readout.
<code>v165_base.py</code>	Sampling, multiscale evolution blocks and recurrent memory.
<code>manifest.json</code>	Released grids, normalization and checkpoint/source identity.
Saved evaluation arrays	270-storm final metrics; Fung-wong route, fields and member provenance.

Model sources are in `models/trackformer_1_2_field/`; evaluation arrays are in `evaluation/`. This document embeds figure coordinates and needs no external images. The released plotting utility regenerates the figure data from saved arrays; it does not perform new inference.

6 Development evaluation

6.1 Matched cases, different forecast pipelines

The completed comparison uses **1,473 daily issues from 270 storms**, with 20 six-hour leads per issue. Scores average leads within daily issues, issues within storms, then storms equally. The cohort was frozen before inference. Trackformer 1.2 averages 50 causal input perturbations; 1.1 uses its causal route pipeline. Cases and targets match, but inputs differ. This is broader development evidence, not an untouched test or architecture ablation.

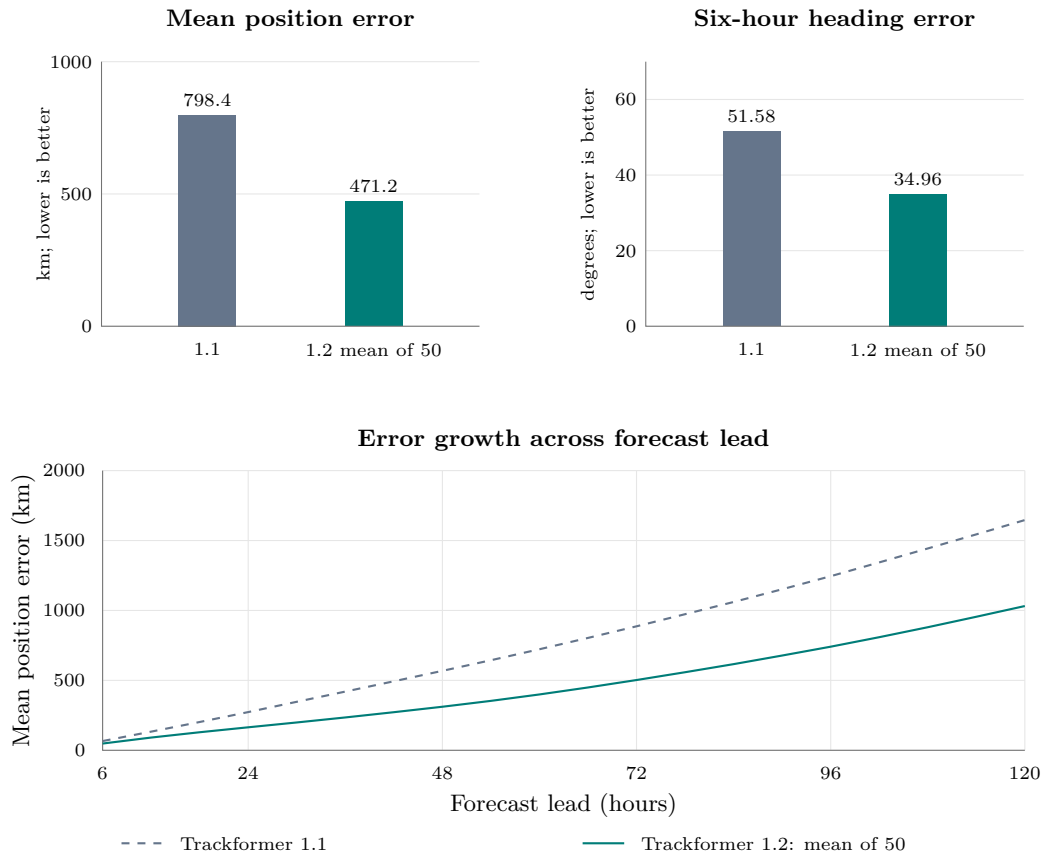


Figure 2. Completed equal-storm development comparison, with every daily issue retained. Positions use a common issue-centred local-km projection. Heading uses common steps where truth and both models move more than 1 km. No uncertainty intervals are claimed.

6.2 Interpretation: alignment, not shape alone

Mean position error improves by 41.0%, from 798.4 to 471.2 km; +120 h error falls from 1,646.4 to 1,031.6 km. At +6 h, error is 48.3 versus 66.1 km. Centred shape similarity is 0.8837 versus 0.7544 for 1.1; normalized Fréchet path similarity is 0.6560 versus 0.5345. These are separate measures, not accuracy percentages or evidence of a win on every storm.

Actual alignment requires predicted and observed positions to coincide at the same valid times. Removing translation and scale can make displaced routes look similar; the shape score cannot replace geographic error or an unshifted overlay. Path similarity responds to displacement but is not a strict same-time distance.

Pressure coverage. The 1.2 central-pressure MAE is 12.62 hPa on 40 storms with valid labels; basin MSLP MAE is 2.72 hPa on 270 storms. No matched 1.1 pressure comparison is available. All 1,473 saved forecast hashes were verified; the earlier 270-case/90-storm study remains separate.

7 Fung-wong: route alignment and intensity

The selected forecast begins at 00 UTC on 7 November 2025 and uses the saved 50-member policy. Figure 3 compares exact six-hour positions without translating, rotating or scaling either route. Great-circle errors are 130.9 km on average, 142.3 km at +120 h and 198.3 km at maximum; all 20 leads are within 200 km. These are separate from the benchmark's local-km scores.

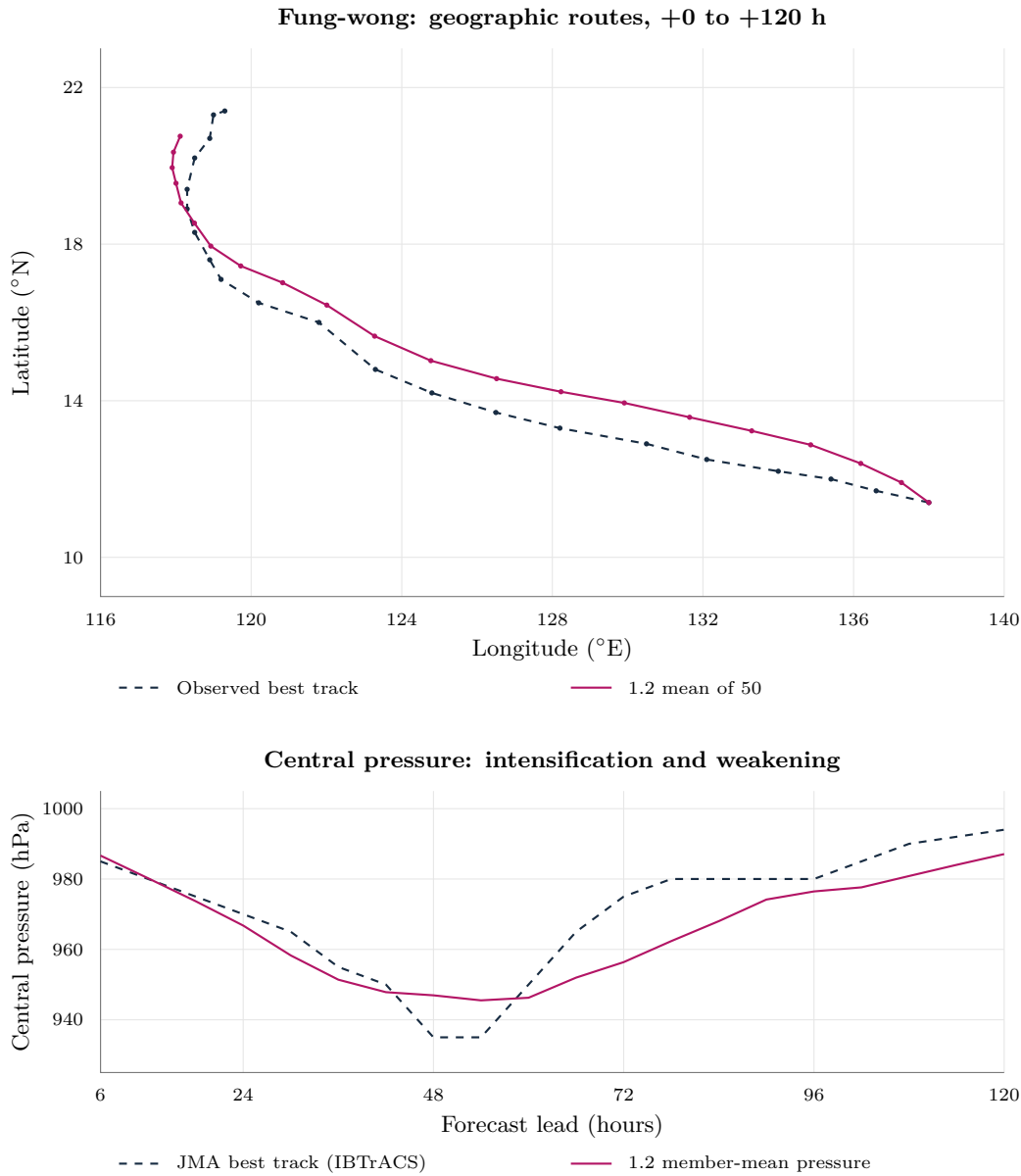


Figure 3. Selected development example, not representative performance. Markers on the route denote matching six-hour valid times, including the issue centre. Observations are verification overlays, never forecast inputs. Pressure MAE is 7.36 hPa over twenty valid labels. The broad track and pressure evolution are similar, but the curves do not perfectly coincide; peak intensity and weakening timing remain imperfect. The companion MP4 shows the actual common-grid model pressure fields and 4-hPa isobars; none of the pressure fields is moved to match observations.

The broad curves align, but position, timing and intensity differences remain. This user-selected development example is not representative performance. The companion MP4 animates the saved fields; no additional inference or route adjustment is used.

8 Pressure-field interpretation and limitations

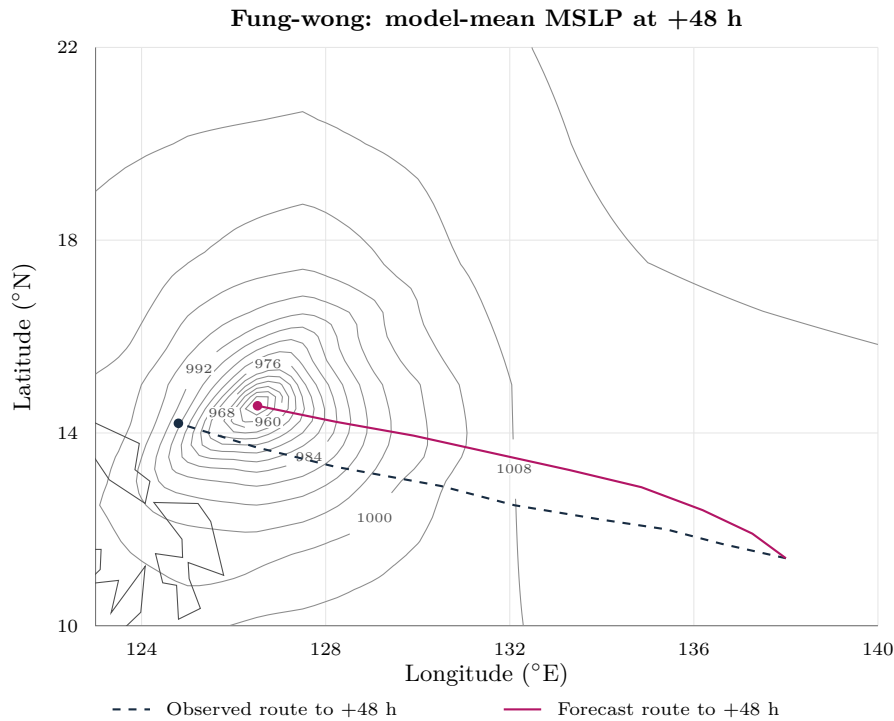


Figure 4. Saved model-mean regional MSLP at +48 h (9 November 2025, 00 UTC), with 4-hPa isobars labelled every 8 hPa. Routes extend only to this valid time. The western boundary is 123°E. The field is not shifted to the observed centre; its minimum need not equal member-mean central pressure.

8.1 Intensity labels and map coverage

Pressure truth is JMA best track from IBTrACS TOKYO_PRES, not a JMA forecast. Native-detail history was unavailable; 0.25° output sampling does not establish native resolution. From +66 h the centre leaves the fixed regional patch. The video then shows only the saved coarse basin field there, while the moving-core readout supplies central pressure. No detailed core is invented beyond coverage.

There is no verified matched 1.1 pressure array for this benchmark. On the repeatedly inspected TIP ten-start diagnostic, 1.2 pressure MAE is worse (30.7 versus 20.1 hPa). Those overlapping starts are not independent tests; route gains do not establish an all-metric improvement.

8.2 What remains unverified

The completed cohort includes 40 recent and 230 historical storms (1980–1999), outside this checkpoint’s fitting and validation years but not certified unused across prior experiments. Complete five-day truth excludes short remaining lifetimes. Historical hindcasts use a model trained on later years and retrospective analyses. Generalization still requires an unused whole-storm holdout, training-only preprocessing and whole-storm uncertainty resampling.

8.3 Conclusion

Trackformer 1.2 links recurrent environmental evolution to a translating pressure anomaly and coupled geographic readout. Its development results improve aggregate route measures over the archived 1.1 pipeline, but leave large absolute errors and intensity failures; the older cohort also showed a short-lead regression. The released model is a reproducible research artifact, not a substitute for official warnings.